

Conrad Borchers

✉ c.borchers@vanderbilt.edu | 🌐 www.cborchers.com

Academic Positions

Vanderbilt University, Peabody College and College of Connected Computing

ASSISTANT PROFESSOR

Nashville, TN, USA

Starting 08/2026

RESEARCH ASSOCIATE

Starting 07/2026

Education

Carnegie Mellon University

Pittsburgh, PA, USA

HUMAN-COMPUTER INTERACTION, PhD

08/2022-06/2026

HUMAN-COMPUTER INTERACTION, MS

08/2022-08/2024

University of Oxford

Oxford, United Kingdom

SOCIAL DATA SCIENCE, MSc

10/2021-08/2022

University of Tübingen

Tübingen, Germany

PSYCHOLOGY, BSc

10/2018-07/2021

Professional Experience

08/2022-06/2026	Graduate Research Assistant , Human-Computer Interaction Institute, Carnegie Mellon University
02/2020-10/2024	Research Affiliate , Lab of Christian Fischer, University of Tübingen
11/2021-07/2022	Student Researcher and Team Lead , Oxford Artificial Intelligence Society
11/2021-07/2022	Graduate Research Assistant , Saïd Business School, University of Oxford
05/2021-09/2021	Research Intern , CAHL Lab (Zachary A. Pardos), University of California, Berkeley
08/2020-01/2021	Undergraduate Research Assistant , Tübingen School of Education
02/2019-07/2020	Undergraduate Research Assistant , Leibniz Knowledge Media Research Center (IWM)

Awards, Fellowships, & Grants

PAPER AND THESIS DISTINCTIONS

2026	Best Full Paper Award [F31] , 16th International Learning Analytics and Knowledge Conference
2025	Best Full Paper Award [F23] , 20th European Conference on Technology Enhanced Learning
2025	Best Long Paper Nomination [F19] , 18th International Conference on Educational Data Mining
2025	Best Short Paper Nomination [S11] , 18th International Conference on Educational Data Mining
2025	Best Short Paper Nomination [S9] , 15h International Learning Analytics and Knowledge Conference
2024	Best Student Paper Award [F13] , International Conference on Quantitative Ethnography
2024	Best Interactive Event Nominee [D1] , 25h International Conference on Artificial Intelligence in Education
2024	Best Student Paper Award [F9] , 17th International Conference on Educational Data Mining
2024	Best Short Paper Award [S7] , 17th International Conference on Educational Data Mining
2024	Best Paper Award (AIS) [F7] , 26th Intl. Conference On Human-Computer Interaction (HCI)
2023	Best Full Paper Honorable Mention [F1] , 13th International Learning Analytics and Knowledge Conference
2022	OII Thesis Prize for Best MSc Dissertation , Oxford Internet Institute
2021	Best Poster Award [S1] , 14th International Conference on Educational Data Mining

REVIEWER AWARDS

- 2025 **Best Reviewer Award**, 12th ACM Conference on Learning@Scale (L@S)
- 2025 **Outstanding Junior Reviewer**, Journal of Learning Analytics

EXTERNAL RESEARCH FUNDING

- Large Grant: Improving Persistence through Adaptive Goal Setting in i-Ready - \$250,000.00**, Digital Promise, the Advancing Innovative Math Solutions (AIMS) Collaboratory, and Learning Data Insights. *Role: PI | Co-PIs: Aleven, Koedinger, Thomas*
- Planning Grant: Improving Persistence through Adaptive Goal Setting in i-Ready - \$10,000.00**, Digital Promise, the Advancing Innovative Math Solutions (AIMS) Collaboratory, and Learning Data Insights. *Role: PI | Co-PIs: Aleven, Koedinger, Thomas*

INTERNAL RESEARCH FUNDING

- 2023 **GuSH Graduate Research Grant (Scaffolding Effects on Problem-Solving in Intelligent Tutoring Systems) - \$750**, Carnegie Mellon University

SELECTIVE FELLOWSHIPS, SCHOLARSHIPS, AND SUMMER SCHOOLS

- 2025 **Siebel Scholars Class of 2026 (\$35,000)**, Thomas and Stacey Siebel Foundation
- 2024 **Ethical AI in Education Fellow**, Learnest
- 2022 **LearnLab Summer School Participant (ITS Track)**, Carnegie Mellon University
- 2022 **Merck Innovation Cup '22 Participant**, Merck KGaA (Merck Group)
- 2021 **Learning Engineering Fellow - 2nd Place in Tools Competition (\$15,000)**, EPIC@Berkeley - UC Berkeley GSE & Schmidt Futures
- 2019/20 **Research Fellow**, European Federation of Psychology Students Associations (EFPSA)
- 2018 **BSc and MSc Scholarship**, German Academic Scholarship Foundation (Studienstiftung)

CONFERENCE FUNDING

- 2025 **Conference Attendance (EDM '25) - \$475**, NSF-SPLICE through the University of Pittsburgh
- 2025 **GSA Conference Funding (LAK '25) - \$750**, Carnegie Mellon University
- 2024 **GSA Conference Funding (LAK '24) - \$750**, Carnegie Mellon University
- 2024 **Full Conference Travel Support (AIED '24) - \$2,000**, Learnest
- 2023 **EDM2023 Travel Scholarship - \$1,250**, International Educational Data Mining Conference
- 2023 **Prof. Ramkumar Memorial Foundation Scholarship - \$100**, International Educational Data Mining Conference
- 2023 **GSA Conference Funding (LAK '23) - \$750**, Carnegie Mellon University

Teaching Experience

TEACHING ASSISTANT

- Fall 2025 **Large Language Models: Methods and Applications**, School of Computer Science, Carnegie Mellon University
- Fall 2023 **User-Centered Research and Evaluation**, School of Computer Science, Carnegie Mellon University

Spring 2023 **Tools for Online Learning**, School of Computer Science, Carnegie Mellon University
Spring 2021 **Cognitive Modeling**, Department of Computer Science, University of Tübingen, Germany

GUEST LECTURES

Summer 2026 **HCII NSF-REU Summer Research Program Seminar**, Guest Lecture: "What Makes Good Research?" hosted by Adam Perer, Carnegie Mellon University
Fall 2025 **CS 565: CS565**, Two Guest Lectures on Publication [W5] and [F20], respectively, hosted by Neil Heffernan, Worcester Polytechnic Institute
Summer 2025 **HCII NSF-REU Summer Research Program Seminar**, Guest Lecture: "What Makes Good Research?" hosted by Laura Dabbish, Carnegie Mellon University
Spring 2025 **Tools for Online Learning**, Guest Lecture on Socio-Emotional Learning, hosted by Steven J. Moore, Carnegie Mellon University
Spring 2025 **CS 5890/6890: Topics in AI and Education**, Guest Lecture: "Enhancing Education with LLMs: Opportunities and Challenges," hosted by Yang Shi, Utah State University
Fall 2024 **CS 568: AI in Adaptive Educational Technologies**, Guest Lecture on Publications [CS7] and [CF9], hosted by Neil Heffernan, Worcester Polytechnic Institute
Spring 2024 **Tools for Online Learning**, Guest Lecture on Socio-Emotional Learning, hosted by Steven J. Moore, Carnegie Mellon University
Summer 2023 **HCII NSF-REU Summer Research Program Seminar**, Guest Lecture: "What Makes Good Research?" hosted by Laura Dabbish, Carnegie Mellon University

WORKSHOPS

April 2026 **Interpreting Tutor Moves with Multimodal Embeddings: A Hands-On Exploration of Clustering and Engagement Signals**, Interactive Workshop on AI Annotation and Teacher Analytics, Learning Analytics and Knowledge (LAK) Conference, Bergen, Norway
Fall 2024 **Multimodal Machine Learning: An Introductory Workshop**, Hosted by Joshua Rosenberg at the University of Tennessee, Knoxville; GitHub: <https://tinyurl.com/utk-mmml>
Fall 2022 **Sentence Embeddings for Twitter Research**, Introduction to Data Science Methods in Education, University of Tennessee, Knoxville; GitHub: <https://tinyurl.com/sembutk>

Academic Publications

Total citations: 999 (Google Scholar, 2026-06-08)

h-index: 18 (Google Scholar, 2026-06-08)

i10-index: 36 (Google Scholar, 2026-06-08)

Number of distinct co-authors on listed papers: 211

Full list of citations per paper: <https://scholar.google.com/citations?user=v5yiJVAAAAAJ&hl=en>

ORCID: 0000-0003-3437-8979

Conference Acceptance Rates (2023-2026): LAK: 30%, EC-TEL: 27%, EDM: 25%, AIED: 24%, ISLS: 30%, LatS: 28%

Italicized author names indicate mentees. An asterisk (*) indicates equal contribution.

JOURNAL ARTICLES

J22 - **Borchers, C.** & Pardos, Z. A. (in press). Generalizability and Transferability of Course Workload Models Across Institutions. *Journal of Educational Data Mining*.

- J21 – **Borchers, C.**, Thomas, D. R., Lin, J., & Koedinger, K. R. (2026). Learning-Aware Reliability Estimation for Tutor Skill Assessment Using Large Language Models. *Journal of Learning Analytics*. <https://doi.org/10.18608/jla.2026.9133>.
- J20 – Simon, S., Şener, B., Pifarré Turmo, M., Greisel, M., Čarapina, M., **Borchers, C.**, Shahrokhian, B., Tajik, E., Chernikova, O., Hatun Ataş, A., Hu, X., & Sitjà Martín, M. (2026). Toward a Taxonomy: Insights from the 2025 CSCL Paradigm Building Workshop. *International Journal of Computer-Supported Collaborative Learning*. <https://doi.org/10.1007/s11412-026-09477-8>
- J19 – Zambrano, A. F., Wei, Z., Zhang, J., Baker, R. S., Ocumpaugh, J. L., Barany, A., Liu, X., Zhou, Y., Paquette, L., Ginger, J., & **Borchers, C.** (2026). Data Plus Theory Equals Codebook: Leveraging LLMs for Human-AI Codebook Development. *Journal of Educational Data Mining*. <https://doi.org/10.5281/zenodo.18352290>
- J18 – **Borchers, C.**, Morina, F., Klein, L., & Fischer, C. (2025). Mapping the landscape of educational use of X (Twitter) in Germany: informal teacher learning in online communities of practice. *Journal of New Approaches in Educational Research*. <https://doi.org/10.1007/s44322-025-00047-8>
- J17 – **Borchers, C.** & Braesemann, F. (2025). The innovation dynamics of programming technologies. *Journal of the Royal Society Interface*. <https://doi.org/10.1098/rsif.2025.0166>
- J16 – **Borchers, C.**, Zhang, J., Fleischer, H., Schanze, S., Aleven, V. & Baker, R. S. (2025). Large Language Models Generalize SRL Prediction to New Languages Within But Not Between Domains. *Journal of Educational Data Mining*. <https://doi.org/10.5281/zenodo.17073680>
- J15 – **Borchers, C.** & Pardos, Z. A. (2025). Course Load Analytics Interventions on Higher Education Course Selection: Experimental Evidence. *Journal of Learning Analytics*. <https://doi.org/10.18608/jla.2025.8473>
- J14 – **Borchers, C.**, Fleischer, H., Schanze, S., Scheiter, K., & Aleven, V. (2025). High scaffolding of an unfamiliar strategy improves conceptual learning but reduces enjoyment compared to low scaffolding and strategy freedom. *Computers & Education*, 236, 105364. <https://doi.org/10.1016/j.compedu.2025.105364>
- J13 – **Borchers, C.**, Wang, Y., Hodge, E., & Rosenberg, J. M. (2025). Decoding Sentiment Signals: Lessons from the Political Reception of the Common Core and Next Generation Science Standards. *Educational Researcher*. <https://doi.org/10.3102/0013189X251336206>
- J12 – **Borchers, C.**, Xu, Y., & Pardos, Z. A. (2025). Workload Overload? Late Enrollment Leads to Course Dropout. *Journal of Educational Data Mining*. <https://doi.org/10.5281/zenodo.14907388>
- J11 – **Borchers, C.**, Darriet, C., Rosenberg, J., & López, F. (2025). Dual Intent in Dual-language Programs: Internet Data Mining of School District Communications. *TechTrends*. <https://doi.org/10.1007/s11528-025-01049-1>
- J10 – **Borchers, C.**, Fleischer, H., Yaron, D. J., McLaren, B. M., Scheiter, K., Aleven, V., & Schanze, S. (2025). Problem-Solving Strategies in Stoichiometry Across Two Intelligent Tutoring Systems: A Cross-National Study. *Journal of Science Education and Technology*. <https://doi.org/10.1007/s10956-024-10197-7>
- J9 – Zhang, L., Lin, J., Sabatini, J., **Borchers, C.**, Weitekamp, D., Cao, M., Hollander, J., Hu, X., & Graesser, A. C. (2025). Data Augmentation for Sparse Multidimensional Learning Performance Data Using Generative AI. *IEEE Transactions on Learning Technologies*. <https://doi.org/10.1109/TLT.2025.3526582>
- J8 – Pritchard, C., **Borchers, C.**, Rosenberg, J. M., Fox, A. K., & Stegenga, S. M. (2024). The datafication of student information on X (Twitter). *Computers and Education Open*, 7, 100197. <https://doi.org/10.1016/j.caeo.2024.100197>
- J7 – **Borchers, C.***, Rosenberg, J.*, & Swartzentruber, R. M.* (2023). Facebook Post Data: A Primer for Educational Research. *Educational Technology Research and Development*. <https://doi.org/10.1007/s11423-023-10269-2>
- J6 – **Borchers, C.***, Eder, T. F.*, Richter, J., Keutel, C., Huettig, F., & Scheiter, K. (2023). A Time Slice Analysis of Dentistry Students' Visual Search Strategies and Pupil Dilation during Diagnosing Radiographs. *PloS one*. <https://doi.org/10.1371/journal.pone.0283376>
- J5 – Pardos, Z. A., **Borchers, C.**, & Yu, R. (2023). Credit hours is not enough: Explaining undergraduate perceptions of course workload using LMS records. *The Internet and Higher Education*, 56, 100882. <https://doi.org/10.1016/j.iheduc.2022.100882>
- J4 – Rosenberg, J., **Borchers, C.**, Stegenga, S., Burchfield, M., Anderson, D., & Fischer, C. (2022). How educational institutions reveal students' personally identifiable information on Facebook. *Learning, Media and Technology*. Advanced Online Publication. <https://doi.org/10.1080/17439884.2022.2140672>

- J3 – Rosenberg, J., **Borchers, C.**, Burchfield, M., Anderson, D., Stegenga, S., & Fischer, C. (2022). Posts About Students on Facebook: A Data Ethics Perspective. *Educational Researcher*. Advanced Online Publication. <https://doi.org/10.3102/0013189X221120538>
- J2 – Rosenberg, J., Burchfield, M., **Borchers, C.**, Gibbons, B., Anderson, D., & Fischer, C. (2021). Social media and students' privacy: What schools and districts should know. *Phi Delta Kappan*, 103(2), 49-53. <https://doi.org/10.1177/00317217211051145>
- J1 – Rosenberg, J., **Borchers, C.**, Dyer, E., Anderson, D. & Fischer, C. (2021). Understanding Public Sentiment About Educational Reforms: The Next Generation Science Standards on Twitter. *AERA Open*, 7(1), 1-17. <https://doi.org/10.1177/23328584211024261>

PEER-REVIEWED CONFERENCE PROCEEDINGS (FULL PAPERS)

- F48 – Gao, G., Gurung, A., Thomas, D. R., Abijuru Kamikazi, M. C., **Borchers, C.**, Gatz, E., Gupta, S., Fancsali, S., Ritter, S., Branstetter, L., Koedinger, K., & Brunskill, E. (in press). Dosage and Dialogue: What Drives Student Math Progress in One-on-One Human-AI Hybrid Tutoring? *Proceedings of the 21st European Conference on Technology Enhanced Learning (EC-TEL 2026)*. Valencia, Spain.
- F47 – Thomas, D. R., Abijuru Kamikazi, M. C., Brandt, C., **Borchers, C.**, & Koedinger, K. R. (in press). An AI-driven and Closed-loop Human Tutor Training and Real-life Assessment System. *Proceedings of the 21st European Conference on Technology Enhanced Learning (EC-TEL 2026)*. Valencia, Spain.
- F46 – Cen, X., Alevén, V., Koedinger, K. R., **Borchers, C.**, & Carvalho, P. F. (in press). Self-Efficacy and Favorability Shape Learning from Tutoring Systems and Paper Practice. *Proceedings of the 21st European Conference on Technology Enhanced Learning (EC-TEL 2026)*. Valencia, Spain.
- F45 – Lee, J., **Borchers, C.**, Alvero, A., Joachims, T., & Kizilcec, R. F. (in press). The Digital Divide in Generative AI: Evidence from Large Language Model Use in College Admissions Essays. *Proceedings of the Thirteenth (2026) ACM Conference on Learning@Scale (L@S)*. Seoul, Republic of Korea. <https://doi.org/10.1145/3774398.3811620>
- F44 – **Borchers, C.**, Viberg, O., & Kizilcec, R. F. (in press). Who Decides in AI-Mediated Learning? The Agency Allocation Framework. *Proceedings of the Thirteenth (2026) ACM Conference on Learning@Scale (L@S)*. Seoul, Republic of Korea. <https://doi.org/10.1145/3774398.3811627>
- F43 – **Borchers, C.**, Zhang, L., Yang, K., Nagashima, T., & Domingue, B. W. (in press). Understanding Student Effort Using Reaction Time Propensities During Problem Solving at Scale. *Proceedings of the Thirteenth (2026) ACM Conference on Learning@Scale (L@S)*. Seoul, Republic of Korea. <https://doi.org/10.1145/3774398.3811624>
- F42 – Kim, J., Li, Q., **Borchers, C.**, Jiang, Z., & Xu, D. (in press). Student, Course Design, or Context? Studying the Determinants of Academic Procrastination at Scale. *Proceedings of the Thirteenth (2026) ACM Conference on Learning@Scale (L@S)*. Seoul, Republic of Korea. <https://doi.org/10.1145/3774398.3811614>
- F41 – **Borchers, C.**, Shahrokhian, B., Balzan, F., Tajik, E., Sankaranarayanan, S., & Simon, S. (in press). Temperature and Persona Shape LLM Agent Consensus With Minimal Accuracy Gains in Qualitative Coding. *Proceedings of the 19th International Conference on Educational Data Mining (EDM)*. Seoul, Republic of Korea.
- F40 – Qiu, E. S., Thomas, D. R., Guo, B., Alevén, V., & **Borchers, C.** (in press). From Heuristics to Analytics: Forecasting Effort and Progress in Online Learning. *Proceedings of the 19th International Conference on Educational Data Mining (EDM)*. Seoul, Republic of Korea.
- F39 – **Borchers, C.**, Rodrigues, L., Torrezão da Costa, N., Xavier, C., & Mello, R. F. (in press). Understanding Teacher Revisions of Large Language Model-Generated Feedback. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, Republic of Korea.
- F38 – **Borchers, C.**, Švábenský, V., Kafle, S. K., Tang, K. K., & Vykopal, J. (in press). Multimodal Analytics of Cybersecurity Crisis Preparation Exercises: What Predicts Success? *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, Republic of Korea.
- F37 – Thomas, D. R., **Borchers, C.**, Vanacore, K. P., Koedinger, K. R., & Kizilcec, R. F. (in press). Modernizing Ground Truth: Four Shifts Toward Improving Reliability and Validity in AI in Education. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, Republic of Korea. Seoul, South Korea.
- F36 – Du, Y., **Borchers, C.**, & Cukurova, M. (in press). Benchmarking Educational LLMs with Analytics: A Case Study on Gender Bias in Feedback. *Proceedings of 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, South Korea.

- F35 – Jin, Q., **Borchers, C.**, Gurung, A., Jackson, S., Agarwal, S., Wang, J., Yu, A., Maheshwary, P., & Aleven, V. (2026). Sticky Help, Fleeting Effects: Session-by-Session Analytics of Teacher Interventions in K-12 Classrooms. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785128>
- F34 – Švábenský, V.*, **Borchers, C.***, Fortuna, E. Cloude, E. B., & Gašević, D. (2026). Mapping 15 Years of Learning Analytics Research: Global Trends, Topics, and Collaboration. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785131>
- F33 – **Borchers, C.**, Deininger, H., & Pardos, Z. A. (2026). Toward Trait-Aware Learning Analytics. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785131>
- F32 – **Borchers, C.**, Gurung, A., Liu, Q., Thomas, D. R., Khalil, M., & Koedinger, K. R. (2026). Brief but Impactful: How Human Tutoring Interactions Shape Engagement in Online Learning. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785041>
- F31 – Liu, Q., Li, L., Švábenský, V., **Borchers, C.**, & Khalil, M. (2026). Measuring the Impact of Student Gaming Behaviors on Learner Modeling. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785036>
- F30 – Zhang, J., **Borchers, C.**, Wang, C., Kumar, V., Teffera, L., McLaren, B. M., & Baker, R. S. (2026). Understanding Gaming the System through Analyzing Self-Regulated Learning in Think-Aloud Protocols. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785025>
- F29 – Thomas, D. R., **Borchers, C.**, & Koedinger, K. R. (2025). Beyond Agreement: Rethinking Ground Truth in Educational AI Annotation. *NCME AIME-Con 2025*, Pittsburgh, PA, USA. <https://doi.org/10.48550/arXiv.2508.00143>
- F28 – Thomas, D. R., **Borchers, C.**, Bhushan, S., Gatz, E., Gupta, S., & Koedinger, K. R. (2025). LLM-Generated Feedback Supports Learning When Learners Choose to Use It. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_33
- F27 – Bhushan, S., Thomas, D. R., **Borchers, C.**, Raghuvanshi, I., Abboud, R., Gatz, E., Gupta, S., & Koedinger, K. R. (2025). Detecting LLM-Generated Short Answers and Effects on Learner Performance. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_4
- F26 – **Borchers, C.**, Nguyen, H. T., Carvalho, P. F., Koedinger, K. R., & Aleven, V. (2025). Goal Setting Engages More Caregivers Than Instructional Support in Online Math Homework. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_5
- F25 – **Borchers, C.**, Nagashima, T., Liu, P., Alibali, M. W., & Aleven, V. (2025). Is More Gamification Better? Evaluating Playful Interactions and Narratives for Algebra Learning. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_6
- F24 – Xia, M.*, Schmucker, R.*, **Borchers, C.***, & Aleven, V. (2025). Optimizing Mastery Learning by Fast-Forwarding Over-Practice Steps. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_37
- F23 – Teich, K., **Borchers, C.**, & Rummel, N. (2025). Using Log Data to Analyze the Impact of Adaptive Support on Self-Regulated Learning in Adult Online Education. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03870-8_31
- F22 – Simon, S., Tajik, E., **Borchers, C.**, Shahrokhian, B., Sankaranarayanan, S., Balzan, F., Strauss, S., Viswanathan, S. A., Atas, A. H., Carapina, M., Liang, L., & Celik, B. (2025). Comparing a Human's and a Multi-Agent System's Thematic Analysis: Assessing Qualitative Coding Consistency. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98420-4_5
- F21 – **Borchers, C.**, Houk, A., Aleven, V., & Koedinger, K. R. (2025). Engagement and Learning Benefits of Goal Setting with Rewards in Human-AI Tutoring. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98459-4_4
- F20 – **Borchers, C.** & Shou, T. (2025). Can Large Language Models Match Tutoring System Adaptivity? A Benchmarking Study. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98417-4_29

- F19 – Gurung, A., Lin, J., Huang, Z., **Borchers, C.**, Baker, R. S., Aleven, V., & Koedinger, K. R. (2025). Starting Seatwork Earlier as a Valid Measure of Student Engagement. *Proceedings of the 18th International Conference on Educational Data Mining (EDM)*. Palermo, Italy. <https://doi.org/10.5281/zenodo.15870183>
- F18 – Simon, S., **Borchers, C.**, Ataş, A. H., Tajik, E., Çelik, B., Čarapina, M., Liu, Y., Shahrokhian, B., Sankaranarayanan, S., Balzan, F., Molinari, G., Jagušt, T., & Liang, L. (2025). Shared Conceptual Ground in CSCL: Evidence From a Global Survey. In Oshima, J., Chen, B., Vogel, F., & Järvelä, J. (Eds.), *Proceedings of the 18th International Conference on Computer-Supported Collaborative Learning - CSCL 2025*. <https://doi.org/10.22318/csc12025.987431>
- F17 – Thomas, D. R., **Borchers, C.**, Kakarla, S., Lin, J., Bhushan, S., Guo, B., Gatz, E., & Koedinger, K. R. (2025). Do Tutors Learn from Equity Training and Can Generative AI Assess It? *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706531>
- F16 – Thomas, D. R., **Borchers, C.**, Kakarla, S., Lin, J., Bhushan, S., Guo, B., Gatz, E., & Koedinger, K. R. (2025). Does Multiple Choice Have a Future in the Age of Generative AI? A Posttest-Only RCT. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706530>
- F15 – Venugopalan, D., Yan, Z., **Borchers, C.**, Lin, J., & Aleven, V. (2025). Combining Large Language Models with Tutoring System Intelligence: A Case Study in Caregiver Homework Support. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706516>
- F14 – Švábenský, V., **Borchers, C.**, Cloude, E. B., & Shimada, A. (2025). Evaluating the Impact of Data Augmentation on Predictive Model Performance. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706485>
- F13 – Zhang, J.*, **Borchers, C.***, & Barany, A. (2024). Studying the Interplay of Self-Regulated Learning Cycles and Scaffolding Through Ordered Network Analysis Across Three Tutoring Systems. *Proceedings of the International Conference on Quantitative Ethnography (ICQE)*, Philadelphia, PA, USA. https://doi.org/10.1007/978-3-031-76335-9_17
- F12 – Yang, K. B.*, **Borchers, C.***, Falhs, A.-C., Echeverria, V., Karumbaiah, S., Rummel, N., & Aleven, V. (2024). Leveraging Multimodal Classroom Data for Teacher Reflection: Teachers' Preferences, Practices, and Privacy Considerations. *Proceedings of the 19th European Conference on Technology Enhanced Learning (EC-TEL)*, Krems, Austria. https://doi.org/10.1007/978-3-031-72315-5_34
- F11 – Baucks, F*, Schmucker, R*, **Borchers, C.**, Pardos, Z. A., & Wiskott, L. (2024). Gaining Insights into Group-Level Course Difficulty via Differential Course Functioning. *Proceedings of the Eleventh (2024) ACM Conference on Learning@Scale (L@S)*. Atlanta, GA, USA. <https://doi.org/10.1145/3657604.3662028>
- F10 – Zhang, J., **Borchers, C.**, Aleven, V., & Baker, R. S. (2024). Using Large Language Models to Detect Self-Regulated Learning in Think-Aloud Protocols. *Proceedings of the 17th International Conference on Educational Data Mining (EDM)*. Atlanta, GA, USA. <https://doi.org/10.5281/zenodo.12729790>
- F9 – **Borchers, C.**, Yang, K., Lin, J., Rummel, N., Koedinger, K. R., & Aleven, V. (2024). Combining Dialog Acts and Skill Modeling: What Chat Interactions Enhance Learning Rates During AI-Supported Peer Tutoring? *Proceedings of the 17th International Conference on Educational Data Mining (EDM)*. Atlanta, GA, USA. <https://doi.org/10.5281/zenodo.12729784>
- F8 – **Borchers, C.**, Liu, X., Lee, H. H., & Zhang, J. (2024). Ethical AIED and AIED Ethics: Toward Synergy Between AIED Research and Ethical Frameworks. *Proceedings of 25th International Conference on Artificial Intelligence in Education (AIED) – BlueSky Track*. Recife, Brazil. https://doi.org/10.1007/978-3-031-64315-6_2
- F7 – Zhang, L., Lin, J., **Borchers, C.**, Sabatini, J., Hollander, J., Cao, M., & Hu, X. (2024). Predicting Learning Performance with Large Language Models: A Study in Adult Literacy. *Proceedings of the 26th International Conference On Human-Computer Interaction (HCI)*. Washington, DC, USA. https://doi.org/10.1007/978-3-031-60609-0_24
- F6 – Nguyen, H. T., **Borchers, C.**, Xia, M., & Aleven, V. (2024). Designing Tools for Caregiver Involvement in Intelligent Tutoring Systems for Middle School Mathematics. *Proceedings of the 2024 Annual Meeting of the International Society of the Learning Sciences (ISLS)*. Buffalo, NY, USA. <https://doi.org/10.22318/icls2024.630637>
- F5 – **Borchers, C.**, Wang, Y., Karumbaiah, S., Ashiq, M., Shaffer, D. W., & Aleven, V. (2024). Revealing Networks: Understanding Effective Teacher Practices in AI-Supported Classrooms using Transmodal Ordered Network Analysis. *Proceedings of the 14th International Learning Analytics and Knowledge Conference (LAK)*. Kyoto, Japan. ACM. <https://doi.org/10.1145/3636555.3636892>
- F4 – **Borchers, C.**, Zhang, J., Baker, R. S., & Aleven, V. (2024). Using Think-Aloud Data to Understand Relations between Self-Regulation Cycle Characteristics and Student Performance in Intelligent Tutoring Systems. *Proceedings of the 14th International Learning Analytics and Knowledge Conference (LAK)*. Kyoto, Japan. ACM. <https://doi.org/10.1145/3636555.3636911>

- F3 – **Borchers, C.**, Carvalho, P. F., Xia, M., Liu, P., Koedinger, K. R., & Aleven, V. (2023). What Makes Problem-Solving Practice Effective? Comparing Paper and AI Tutoring. *Proceedings of the 18th European Conference on Technology Enhanced Learning (EC-TEL)*. Aveiro, Portugal. https://doi.org/10.1007/978-3-031-42682-7_4
- F2 – Karumbaiah, S., **Borchers, C.**, Shou, T., Falhs, A.-C., Liu, C., Nagashima, T., Rummel, N., & Aleven, V. (2023). A Spatiotemporal Analysis of Teacher Practices in Supporting Student Learning and Engagement in an AI-enabled Classroom. *Proceedings of the 24th International Conference on Artificial Intelligence in Education (AIED)*. Tokyo, Japan. https://doi.org/10.1007/978-3-031-36272-9_37
- F1 – **Borchers, C.** & Pardos, Z. A. (2023). Insights into undergraduate pathways using course load analytics. *Proceedings of the 13th International Learning Analytics and Knowledge Conference (LAK)*. Arlington, TX, USA. ACM. <https://doi.org/10.1145/3576050.3576081>

PEER-REVIEWED CONFERENCE PROCEEDINGS (SHORT/POSTER PAPERS)

- S32 – Yang, K., Pan, Y., **Borchers, C.**, Rummel, N., & Aleven, V. (in press). Supporting Teacher Reflection and Instructional Insight through Multimodal Learning Analytics: An Exploratory Classroom Study with Reflecto. *Proceedings of the 21st European Conference on Technology Enhanced Learning (EC-TEL 2026)*. Valencia, Spain.
- S31 – Pankiewicz, M., **Borchers, C.**, & Baker, R. S. (in press). How Students Engage with LLM Feedback Matters: Feedback Processing Time and Learning in Introductory Programming. *Proceedings of the 21st European Conference on Technology Enhanced Learning (EC-TEL 2026)*. Valencia, Spain.
- S30 – **Borchers, C.** (in press). Framing the 5% Problem: Teachers' Perspectives on Persistence in Educational Technology. *Proceedings of the 21st European Conference on Technology Enhanced Learning (EC-TEL 2026)*. Valencia, Spain.
- S29 – **Borchers, C.**, Schmucker, R., Tiwari, A., & Pardos, Z. A. (in press). Push and Pull in Community College Cross-Enrollment: Remoteness, Articulation, and Student Mobility. *Proceedings of the Thirteenth (2026) ACM Conference on Learning@Scale (L@S)*. Seoul, Republic of Korea. <https://doi.org/10.1145/3774398.3811580>
- S28 – Noh, H., Chowdhary, A., Ooge, J., Aleven, V., & **Borchers, C.** (in press). Simulating Learners' Task-Selection Strategies and System Constraints in Mastery Learning. *Proceedings of the 19th International Conference on Educational Data Mining (EDM)*. Seoul, Republic of Korea.
- S27 – **Borchers, C.**, Zhang, J., & Gurung, A. (in press). Representation Learning to Study Temporal Dynamics in Tutorial Scaffolding. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, Republic of Korea. Seoul, South Korea.
- S26 – Lyu, Q*, **Borchers, C.***, Xia, M., Xiao, K., Carvalho, P. F., Koedinger, K. R., & Aleven, V. (in press). Evaluating a Data-Driven Redesign Process for Intelligent Tutoring Systems. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, Republic of Korea. Seoul, South Korea.
- S25 – Huang, X., **Borchers, C.**, Zhang, J., & Lajoie, S. P. (in press). Physiological and Semantic Patterns in Medical Teams Using an Intelligent Tutoring System. *Proceedings of the 27th International Conference on Artificial Intelligence in Education (AIED 2026)*. Seoul, Republic of Korea. Seoul, South Korea.
- S24 – Tajik, E., **Borchers, C.***, Shahrokhian, B*, Simon, S., Keramati, A., Pal, S., & Sankaranarayanan, S. (2026). Disagreement as Data: Reasoning Trace Analytics in Multi-Agent Systems. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785101>
- S23 – Zhang, J., **Borchers, C.**, Cohn, C., Srivastava, N., Snyder, C., Guo, S., Ashwin, T. S., Mohammed, N., Noh, H., & Biswas, G. (2026). Using Large Language Models to Detect Socially Shared Regulation of Learning in Collaborative Learning. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785083>
- S22 – **Borchers, C.**, Patel, M., Lee, S., & Botelho, A. F. (2026). Disentangling Learning from Judgment: Representation Learning for Open Response Analytics. *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK)*. Bergen, Norway. ACM. <https://doi.org/10.1145/3785022.3785042>
- S21 – Thomas, D. R., **Borchers, C.**, Lin, J., Kakarla, S., Bhushan, S., Gatz, E., Gupta, S., Abboud, R., & Koedinger, K. R. (2025). Leveraging LLMs to Assess Tutor Moves in Real-Life Dialogues: A Feasibility Study. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03873-9_35
- S20 – Falhs, A.-C., **Borchers, C.**, Echeverria, V., Yang, K., Rummel, N., & Aleven, V. (2025). How Expertise Levels Shape Preferences and Reflection Needs: Towards AI Reflection Systems for Teacher Empowerment. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03873-9_11

- S19 – Fleischer, H., **Borchers, C.**, Schanze, S., & Aleven, V. (2025). Error Classification in Stoichiometry Tutoring Systems with Different Levels of Scaffolding: Comparing Rule-Based Classification and Machine Learning. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03873-9_12
- S18 – **Borchers, C.**, Sheel, S., Pai, A., Shah, S., & Pardos, Z. A. (2025). Designing the Course Load Analytics Platform. *Proceedings of the 20th European Conference on Technology Enhanced Learning (EC-TEL)*, Newcastle and Durham, UK. https://doi.org/10.1007/978-3-032-03873-9_10
- S17 – Chen, E., Tang, X., Xi, A., Lin, C., **Borchers, C.**, Gupta, S., Lin, J., & Koedinger, K. R. (2025). VTutor for High-Impact Tutoring at Scale: Managing Engagement and Real-Time Multi-Screen Monitoring with P2P Connections. *Proceedings of the 12th ACM Conference on Learning at Scale (L@S)*, Palermo, Italy. <https://doi.org/10.1145/3698205.3733948>
- S16 – Fleischer, H., Noglik, A., **Borchers, C.**, & Schanze, S. (2025). Does Student Learning Rate Depend on Feedback Type and Prior Knowledge? *Proceedings of the 18th International Conference on Educational Data Mining (EDM 2025)*. Palermo, Italy. <https://doi.org/10.5281/zenodo.15870227>
- S15 – Jin, Q., **Borchers, C.**, Fancsali, S., & Aleven, V. (2025). Who to Help? A Time-Slice Analysis of K-12 Teachers' Decisions in Classes with AI-Supported Tutoring. *Proceedings of the 18th International Conference on Educational Data Mining (EDM 2025)*. Palermo, Italy. <https://doi.org/10.5281/zenodo.15870229>
- S14 – **Borchers, C.**, Nguyen, H. T., Carvalho, P. F., Koedinger, K. R., & Aleven, V. (2025). Involving Parents in Tutoring Systems to Increase Content Confidence: A Design Probe Study *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98465-5_14
- S13 – **Borchers, C.**, Peng, C., Lyu, Q., Carvalho, P. F., Koedinger, K. R., & Aleven, V. (2025). Student Perceptions of Adaptive Goal Setting Recommendations: A Design Prototyping Study. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98462-4_31
- S12 – Thomas, D. R., **Borchers, C.**, Bhushan, S., Kakarla, S., Houk, A., Abboud, R., Gatz, E., Gupta, S., & Koedinger, K. R. (2025). Improving Open-Response Assessment with LearnLM. *Proceedings of 26th International Conference on Artificial Intelligence in Education (AIED)*. Palermo, Italy. https://doi.org/10.1007/978-3-031-98462-4_28
- S11 – **Borchers, C.** (2025). Toward Sufficient Statistical Power in Algorithmic Bias Assessment: A Test for ABROCA. *Proceedings of the 18th International Conference on Educational Data Mining (EDM)*. Palermo, Italy. <https://doi.org/10.5281/zenodo.15870256>
- S10 – Butler, R., **Borchers, C.**, Asher, M., Lee, Y., Karnataki, S., Dangi, S., Athreya, S., Stamper, J., Ogan, A., & Carvalho, P. (2025). Does the Doer Effect Generalize to Non-WEIRD Populations? Toward Analytics in Radio and Phone-Based Learning. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706505>
- S9 – **Borchers, C.** & Baker, R. S. (2025). ABROCA Distributions for Algorithmic Bias Assessment: Considerations Around Interpretation. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706498>
- S8 – **Borchers, C.***, Ooge, J.*, Peng, C., & Aleven, V. (2025). How Learner Control and Explainable Learning Analytics About Skill Mastery Shape Student Desires to Finish and Avoid Loss in Tutored Practice. *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK)*. Dublin, Ireland. ACM. <https://doi.org/10.1145/3706468.3706484>
- S7 – **Borchers, C.**, Xu, Y., & Pardos, Z. A. (2024). Are You an Early Dropper or Late Shopper? Mining Enrollment Transaction Data to Study Procrastination in Higher Education. *Proceedings of the 17th International Conference on Educational Data Mining (EDM)*. Atlanta, GA, USA. <https://doi.org/10.5281/zenodo.12729852>
- S6 – Peng, C., **Borchers, C.**, & Aleven, V. (2024). Designing Homework Support Tools for Middle School Mathematics Using Intelligent Tutoring Systems. *Proceedings of the 2024 Annual Meeting of the International Society of the Learning Sciences (ISLS)*. Buffalo, NY, USA. <https://doi.org/10.22318/icls2024.989202>
- S5 – Shou, T., **Borchers, C.**, Karumbaiah, S., & Aleven, V. (2023). Optimizing Parameters for Accurate Position Data Mining in Diverse Classrooms Layouts. *Proceedings of the 16th International Conference on Educational Data Mining (EDM)*. Bengaluru, India. <https://doi.org/10.5281/zenodo.8115685>
- S4 – **Borchers, C.**, Klein, L., Johnson, H., & Fischer, C. (2023). Timing Matters: Inferring Educational Twitter Community Switching from Membership Characteristics. *Proceedings of the 16th International Conference on Educational Data Mining (EDM)*. Bengaluru, India. <https://doi.org/10.5281/zenodo.8115752>

- S3 – Karumbaiah, S.*, **Borchers, C.***, Falhs, A.-C.*, Holstein, K., Rummel, N., & Aleven, V. (2023). Teacher Noticing and Student Learning in Human-AI Partnered Classrooms: A Multimodal Analysis. *Proceedings of the 2023 Annual Meeting of the International Society of the Learning Sciences (ISLS)*. Montréal, Canada. <https://doi.org/10.22318/icls2023.151200>
- S2 – **Borchers, C.**, Rosenberg, J., Gibbons, B., Burchfield, M., & Fischer, C. (2021). To Scale or Not to Scale: Comparing Popular Sentiment Analysis Dictionaries on Educational Twitter Data. *Proceedings of the 14th International Conference on Educational Data Mining (EDM)*. Paris, France.
- S1 – Burchfield, M., Rosenberg, J., **Borchers, C.**, Thomas, T., Gibbons, B., & Fischer, C. (2021). Are Violations of Student Privacy “Quick and Easy”? Investigating the Privacy of Students’ Images and Names in the Context of K-12 Educational Institution’s Posts on Facebook. *Proceedings of the 14th International Conference on Educational Data Mining (EDM)*. Paris, France.

PEER-REVIEWED WORKSHOP AND DOCTORAL CONSORTIUM PUBLICATIONS

- W9 – **Borchers, C.**, Koedinger, K. R., & Aleven, V. (2025). Intelligent Support for Practice Goal Setting to Enhance Learning. *Doctoral Consortium of the 26th International Conference on Artificial Intelligence in Education (AIED)*. https://doi.org/10.1007/978-3-031-99261-2_43
- W8 – **Borchers, C.***, Tian, X.*, Boyer, K. E., Israel, M. (2025). Combining Log Data and Collaborative Dialogue Features to Predict Project Quality in Middle School AI Education. *9th Educational Data Mining in Computer Science Education (CSEDM) Workshop*.
- W7 – Sankaranarayanan, S., **Borchers, C.**, Simon, S., Tajik, E., Atas, A. H., Celik, B., Balzan, F., & Shahrokhian, B. (2025). Automating Thematic Analysis with Multi-Agent LLM Systems. *From Data to Discovery: LLMs for Qualitative Analysis in Education at LAK25*.
- W6 – **Borchers, C.**, Thomas, D. R., Lin, J., Abboud, R., & Koedinger, K. R. (2025). Augmenting Human-Annotated Training Data with Large Language Model Generation and Distillation in Open-Response Assessment. *Proceedings of the Second Workshop on Generative AI for Learning Analytics (GenAI-LA) at LAK25*.
- W5 – Kakarla, S., **Borchers, C.**, Thomas, D. R., Bhushan, S., & Koedinger, K. R. (2025). Comparing Few-Shot Prompting of GPT-4 LLMs with BERT Classifiers for Open-Response Assessment in Tutor Equity Training. *Proceedings of the Innovation and Responsibility in AI-Supported Education (iRAISE) Workshop at AAAI25*.
- W4 – Aleven, V., **Borchers, C.**, Huang, Y., Nagashima, T., McLaren, B., Carvalho, P., Popescu, O., Sewall, J., & Koedinger, K. (2024). An Integrated Platform for Studying Learning with Intelligent Tutoring Systems: CTAT+TutorShop. *Proceedings of the Fifth Annual Workshop on Learning@Scale 2024: A/B Testing and Platform-Enabled Learning Research*.
- W3 – Zhang, L., Lin, J., **Borchers, C.**, Cao, M., & Hu, X. (2024). 3DG: A Framework for Using Generative AI for Handling Sparse Learner Performance Data From Intelligent Tutoring Systems. *Proceedings of the First Workshop on Generative AI for Learning Analytics (GenAI-LA) at LAK24*.
- W2 – Han, Z. F., Lin, J., Gurung, A., Thomas, D. R., Chen, E., **Borchers, C.**, Gupta, S., & Koedinger, K. R. (2024). Improving Assessment of Tutoring Practices using Retrieval-Augmented Generation. *Proceedings of the AAAI2024 Workshop on AI for Education - Bridging Innovation and Responsibility*.
- W1 – **Borchers, C.***, Gala, D. S.*, Gilbert, B.*, Oravkin, E., Bounsi, W., Asano, Y. M., & Kirk, H. R. (2022). Looking for a Handsome Carpenter! Debiasing GPT-3 Job Advertisements. *Proceedings of the 4th Workshop on Gender Bias in Natural Language Processing (NAACL)*.

PEER-REVIEWED CONFERENCE DEMO PUBLICATIONS

- D3 – Chen, E., Tang, X., Xi, A., Lin, C., **Borchers, C.**, Gupta, S., Lin, J., & Koedinger, K. R. (2025). Demo of VTutor for High-Impact Tutoring at Scale: A Real-Time Multi-Screen Tutor Support System with P2P Connections. *Proceedings of the 12th ACM Conference on Learning at Scale (L@S)*, Palermo, Italy.
- D2 – Yang, K., Zhao, Y., **Borchers, C.**, Fang, Y., Kim, J., Butt, A., Popescu, O., Gupte, S., Chiu, A., Shen, P., Rummel, N., & Aleven, V. (2025). Reflecto: A Teacher Reflection Tool Leveraging Multimodal Learning and Teaching Analytics. *26th International Conference on Artificial Intelligence in Education (AIED)*.
- D1 – Rao, J., **Borchers, C.**, & Lin, J. (2024). Coursera-REC: Explainable MOOCs Course Recommendation using RAG-facilitated LLMs. *25th International Conference on Artificial Intelligence in Education (AIED)*.

Presentations & Invited Talks

CONFERENCE PRESENTATIONS

- P11 – Thomas, D. R., **Borchers, C.**, Vanacore, K., Whitmer, J., Koedinger, K. R., & Kizilcec, R. F. (2026). Consensus is Not Correctness: Rethinking Reliability in Educational AI. Presentation at the Stanford Education Data Science Conference 2026, Stanford, CA.
- P10 – Lee, J., **Borchers, C.**, Alvero, A., Joachims, T., & Kizilcec, R. F. (2026). The Digital Divide with Generative AI: Examining LLM Adoption and Lexical Convergence Across Socioeconomic Groups in College Admissions Essays. Presentation at the Stanford Education Data Science Conference 2026, Stanford, CA.
- P9 – Fleischer, H., **Borchers, C.**, Schanze, S., & Aleven, V. (2024, September). Fehlerklassifizierung beim Tutor-Gestützten Lösen von Stöchiometrieaufgaben [Error Classification in Tutor-Supported Stoichiometry Problem Solving]. Presentation at the 2024 Annual Meeting of the Gesellschaft für Didaktik der Chemie und Physik (GDChP), Bochum, Germany.
- P8 – Rosenberg, J. M., Pritchard, C., **Borchers, C.**, Burchfield, M., Fischer, C., Fox, A. K., & Stegenga, S. (2024, March). Data ethics in the open: Navigating student information risks in educational social media. Presentation at the Data Science & Social Research, 4th International Conference. <https://dssr2024.unina.it/index.html>
- P7 – Gold, G., **Borchers, C.**, & Carvalho P. F. (2024, March). Students' Academic Performance and Goal Orientation Relate to Initial Knowledge but Not Learning Rate. Poster presented at the 14th International Learning Analytics and Knowledge Conference (LAK), Kyoto, Japan.
- P6 – Fischer, C., Morina, F., **Borchers, C.**, & Klein, L. (2023, August). Online Communities of Practice on Twitter: A Case Study of the German Educational Twittersphere. Paper presented at the 20th Biennial EARLI Conference, Thessaloniki, Greece.
- P5 – **Borchers, C.**, & Braesemann, F.. Correlation networks predict success of digital technologies (2022, November). Paper presented at the 11th International Conference on Complex Networks and their Applications, Palermo, Italy.
- P4 – Fischer, C., Morina, F., **Borchers, C.**, Klein, L., & Eich, M. (2022, April). Informal Professional Development in Online Communities of Practice at Scale: Mapping Nationwide Educational Twitterspheres. Paper presented at the 2022 Annual Meeting of the American Educational Research Association, San Diego, CA.
- P3 – **Borchers, C.**, Eder, T., & Scheiter, K. (2021, September). How Effective is Reflective search? A Time Slice Analysis of Dentistry Students' Visual Search Strategies and Pupil Dilation during Diagnosing Radiographs. Paper presented at the 2nd Workshop on Mental Effort, Virtual Conference.
- P2 – Rosenberg, J., **Borchers, C.**, Gibbons, B., Dyer, E., Anderson, D., & Fischer, C. (2021, April). Don't worry, be happy? A sentiment analysis of the #NGSSchat Twitter community. Paper presented at the 2021 Annual Meeting of the American Educational Research Association, Virtual Conference.
- P1 – Fischer, C., Morina, F., **Borchers, C.**, & Klein, L. (2021, March). Using Big Data to Support Educational Change: Twitter as a Platform for Informal Professional Development. Paper presented at the 2021 Annual Meeting of the Leibniz Education Research Network, Hannover, Germany.

INVITED TALKS

- T20 – **Borchers, C.** (2026, April). Leveraging Learning Process Data to Understand Student Workload, Effort, and Individual Differences. LEAD Retreat, LEAD Graduate School & Research Network, Eberhard Karls Universität Tübingen.
- T19 – **Borchers, C.** (2026, April). Effort and Multimodality as Frontiers of AI + Education Futures. Monday Meeting, Hector AI+Education Future Fund, Eberhard Karls Universität Tübingen.
- T18 – **Borchers, C.** (2026, March). Learning Analytics for Persistence: Designing Effort-Aware AI Systems in K-12 and Higher Education. College of Education and Human Development, Texas A&M University.
- T17 – **Borchers, C.** (2026, February). Modeling Student Effort: Data Science for Persistence, Workload, and Learning. Department of Data Science, New Jersey Institute of Technology.
- T16 – **Borchers, C.** (2026, February). Artificial Intelligence for Modeling Effort, Workload, and Persistence: From K-12 Classrooms to Higher Education Institutions. Peabody College and College of Connected Computing, Vanderbilt University.
- T15 – **Borchers, C.** (2026, February). Data Science Approaches to Modeling and Supporting Student Persistence and Workload. School of Data Science, University of Virginia.
- T14 – **Borchers, C.** (2026, February). Computational Models of Effort, Workload, and Skill Acquisition in Educational Contexts. Department of Psychology, University of Illinois Urbana-Champaign.

- T13 – **Borchers, C.** (2025, December). Using Log Data to Analyze the Impact of Adaptive Support on Self-Regulated Learning in Adult Online Education [F23]. Collaboration Call, AIMS (Advancing Innovative Math Solutions) Collaboratory.
- T12 – **Borchers, C.** (2025, December). Goal Support in Hybrid Tutoring: Impact, Scale, and Lessons Learned. Collaboration Call, Learning Engineering Virtual Institute.
- T11 – **Borchers, C.** (2025, November). Toward Effort-Sensitive AI: Modeling and Supporting Student Persistence Across Contexts. GSE Colloquium Series in Education Data Science, Stanford University.
- T10 – **Borchers, C.** (2025, November). Effort-Sensitive AI for Learning: Closing Persistence Gaps Through Adaptive Design. Invited talk at the Digital Learning Lab (Mark Warschauer), University of California, Irvine.
- T9 – **Borchers, C.** (2025, October). Shaping Self-Regulated Learners in the LLM Era: Evidence, Models, and Design. *Information Science Colloquium*. Ann S. Bowers College of Computing and Information Science, Cornell University.
- T8 – **Borchers, C.** (2025, October). Effort-Sensitive AI for Learning: Measuring and Shaping Persistence. *Learning Sciences & Learning Technologies: Research Directions for the Next Decade*. École polytechnique fédérale de Lausanne (EPFL).
- T7 – **Borchers, C.** (2025, May). Combining LLMs with Tutoring System Intelligence: A Case Study in Caregiver Homework Support. LearnLab Cooperate Partners Summit.
- T6 – **Borchers, C.** (2025, April). Designing Tools for Caregiver Involvement in Intelligent Tutoring Systems. GSA Scholar to Scholar Carnegie Mellon University. **GSA Academic Affairs Member Award 2024/25.**
- T5 – **Borchers, C.** (2024, November). LLMs in Educational Research: Recent Trends and Three Future Directions. Interdisciplinary Education Seminar at the University of Tennessee, Knoxville.
- T4 – **Borchers, C.** (2024, September). To Improve Education with LLMs, Do More Than Just LLMs. Learning Analytics (AMA Series; M.S.Ed. Online), University of Pennsylvania GSE
- T3 – **Borchers, C.** (2024, May). Envisioning LLMs as Tools for Educational Research and Practice Rather than Instructional Replacements. Invited Talk at the ALTTAI Seminar at the University of Memphis.
- T2 – **Borchers, C.** (2022, October). Debiasing GPT-3 Job Ads. The Data Skeptic Podcast hosted by Kyle Polich.
- T1 – **Borchers, C.** (2022, September). Leveraging data science to improve educational processes: From K-12 student privacy on social media to higher education student workload. Invited talk at the Department of Theory & Practice in Teacher Education, University of Tennessee, Knoxville.

INVITED PANELS

- IP5 – Panelist, Curriculum Analytics for 21st-Century Competencies Workshop (2026, March). Learning Analytics and Knowledge Conference (LAK'26).
- IP4 – Generative AI to streamline institutional and system workflows (2026, March). OECD Digital Education Outlook Conference.
- IP3 – Virtual Literacy Panel Series – Technology and Literacy: Building Skills for Learning and Careers (2025, October). Carnegie Library of Pittsburgh hosted by Robert Longo.
- IP2 – Early Career Researchers and AI (2025, October). Empowering Learners in the Age of AI (ELAI) Global Conference 2025.
- IP1 – Early Career Panel (2025, July). 26th International Conference on Artificial Intelligence in Education (AIED).

Mentoring

Summer 2026	Mahika Gunjkar , Research Assistant, Carnegie Mellon University
Summer 2026	Sofia Velasquez Sierra , Research Assistant, Carnegie Mellon University
Summer 2026	Archita Jain , Research Assistant, Carnegie Mellon University
Summer 2026	Veronica Shao , SURA, School of Computer Science, Carnegie Mellon University
Summer 2026	Katherine Chen , SURA, School of Computer Science, Carnegie Mellon University
Summer 2026	Zoe Setiadi , NSF REU, Carnegie Mellon University
Summer 2026	Janet Liu , NSF REU, Carnegie Mellon University
Spring 2026	Adrian (Jisue) Shin , Research Assistant, Carnegie Mellon University

Spring 2026	Aishwarya Chitale , Graduate Research Assistant, Carnegie Mellon University
Spring 2026	Ishan Gandhi , Undergraduate Research Assistant, Carnegie Mellon University
Spring 2026	Carina Zheng , Independent Undergraduate Research, Carnegie Mellon University
Spring 2026	Meenakshi Nair , Independent Undergraduate Research, Carnegie Mellon University
Spring 2026	Sebastien Nyiri , Undergraduate Independent Study, Carnegie Mellon University
2025/26	Kent Grass , Research Assistant, PLUS Tutoring, Carnegie Mellon University
Fall 2025	Aidan Plummer , Undergraduate Independent Study, Carnegie Mellon University
Fall 2025	Ian Youn , Undergraduate Independent Study, Carnegie Mellon University
Summer 2025	Manit Patel , SURA, School of Computer Science, Carnegie Mellon University
Summer 2025	Hana Benko , SURA, School of Computer Science, Carnegie Mellon University
Summer 2025	Aarna Chowdhary , SURA, School of Computer Science, Carnegie Mellon University
Summer 2025	Arush Khare , SURA, School of Computer Science, Carnegie Mellon University
Summer 2025	Haley Noh , NSF REU, Carnegie Mellon University
Summer 2025	Karen Xiao , NSF REU, Carnegie Mellon University
Summer 2025	Shi (Eric) Qiu , NSF REU, Carnegie Mellon University
Spring 2025	Vardaan Gupta , Undergraduate Independent Study, Carnegie Mellon University
Spring 2025	Kevin Tang , Independent Undergraduate Research, Carnegie Mellon University
2024/25	Katherine Zhao , SCS SURA and Lab Intern, Carnegie Mellon University
2024/25	Siyu (Dorian) Pan , Lab Intern, Carnegie Mellon University
Summer 2024	Hrugved Anantrao Pawar , Summer Lab Intern, Carnegie Mellon University
Summer 2024	Kloe Chen , Summer Lab Intern, Carnegie Mellon University
Summer 2024	Doris Gao , Summer Lab Intern, Carnegie Mellon University
Summer 2024	Zoey Yan , NSF REU, Carnegie Mellon University
Summer 2024	Devika Venikopalan , SURA, School of Computer Science, Carnegie Mellon University
Fall 2023	Snigdha Tiwari , Research Assistant, Carnegie Mellon University
Fall 2023	Dhruvisha Mondhe , METALS Independent Study, Carnegie Mellon University
2023/24/25	Kennion Gubler , NSF REU and Independent Study, Carnegie Mellon University
Summer 2023	Ha Tien Nguyen , NSF REU, Carnegie Mellon University
Summer 2023	Cindy Peng , NSF REU, Carnegie Mellon University
Spring 2023	Jialin Ye , MHCI Independent Study, Carnegie Mellon University
Spring 2023	Avani Guduri , Undergraduate Independent Study, Carnegie Mellon University
2022/24/25	Tianze (Steven) Shou , Independent Study, Carnegie Mellon University

Service to the Academic Community

WORKSHOPS ORGANIZED

2026	A CACL Taxonomy Building Workshop , Martin Greisel, Burcu Sener, Sebastian Simon, Conrad Borchers, Elham Tajik and Bahar Shahrokhian	<i>ISLS'26</i>
2025	CACL Paradigm Building Workshop , Conrad Borchers, Francesco Balzan, Berkan Celik, Bahar Shahrokhian, Amine Hatun Atas, Mia Čarapina, Sebastian Simon, Elham Tajik, Gaëlle Molinari and Carol Kwai Kuen Chan	<i>ISLS'25</i>

SYMPOSIA ORGANIZED

2026	Symposium on Fostering Human Agency for AI-Mediated Learning , Planning and Organizing Contributor	<i>KTH</i>
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PEER REVIEW

- 2026 **Telematics and Informatics Reports**, Ad-hoc Reviewer
- 2026 **International Journal of Human-Computer Studies**, Ad-hoc Reviewer
- 2026 **Assessing Writing**, Ad-hoc Reviewer
- 2026 **Array**, Ad-hoc Reviewer
- 2025 **AIS Transactions on Human-Computer Interaction**, Ad-hoc Reviewer
- 2025-2026 **Discover Education**, Ad-hoc Reviewer
- 2025 **PeerJ Computer Science**, Ad-hoc Reviewer
- 2025 **AIED Encyclopedia – Chapter Review**, Ad-hoc Reviewer
- 2025 **Entertainment Computing**, Ad-hoc Reviewer
- 2025 **European Journal of Education**, Ad-hoc Reviewer
- 2025-2026 **Computers & Education: Artificial Intelligence**, Ad-hoc Reviewer
- 2025 **Acta Psychologica**, Ad-hoc Reviewer
- 2025-2026 **Journal of Computer Assisted Learning**, Ad-hoc Reviewer
- 2025 **PLOS One**, Ad-hoc Reviewer
- 2025 **Information Processing and Management**, Ad-hoc Reviewer
- 2025 **Computers in Human Behavior Reports**, Ad-hoc Reviewer
- 2025 **IEEE Transactions on Knowledge and Data Engineering**, Ad-hoc Reviewer
- 2025-2026 **Computers & Education**, Ad-hoc Reviewer
- 2025-2026 **Social Sciences & Humanities Open**, Ad-hoc Reviewer
- 2025 **Studies in Educational Evaluation**, Ad-hoc Reviewer
- 2025 **Computers in Human Behavior: Artificial Humans**, Ad-hoc Reviewer
- 2025 **Cogent Education**, Ad-hoc Reviewer
- 2025 **Journal of Science Education and Technology**, Ad-hoc Reviewer
- 2025 **International Journal of Adolescence and Youth**, Ad-hoc Reviewer
- 2025 **BMC Medical Education**, Ad-hoc Reviewer
- 2025 **International Journal of Human-Computer Interaction**, Ad-hoc Reviewer
- 2025-2026 **Journal of Educational Data Mining**, Ad-hoc Reviewer
- 2024-2025 **IEEE Transactions on Learning Technologies**, Ad-hoc Reviewer
- 2024 **Frontiers in Artificial Intelligence**, Ad-hoc Reviewer
- 2024 **Springer Scientific Reports**, Ad-hoc Reviewer
- 2024-2025 **British Journal of Educational Technology**, Ad-hoc Reviewer
- 2024 **Frontiers in Education**, Ad-hoc Reviewer
- 2024-2026 **Journal of Learning Analytics**, Ad-hoc Reviewer
- 2024 **Journal of Infrastructure, Policy and Development**, Ad-hoc Reviewer
- 2024-2026 **Teaching and Teacher Education**, Ad-hoc Reviewer
- 2024 **American Journal of Education**, Ad-hoc Reviewer
- 2025 **ICIS**, Conference Proceedings
- 2024 **RecSys**, Conference Proceedings
- 2024-2026 **L@S**, Conference Proceedings
- 2024 **CSCW**, Conference Proceedings
- 2024-2026 **EDM**, Conference Proceedings
- 2024-2026 **AIED**, Conference Proceedings
- 2024-2026 **CHI**, Conference Proceedings
- 2024-2026 **LAK**, Conference Proceedings
- 2024 **AAAI**, Workshop Track
- 2023-2026 **ISLS**, Conference Proceedings
- 2022-2024 **AERA**, Annual Meeting Proceedings
- 2023 **NeurIPS**, Workshop Track

EDITORIAL POSITIONS

2025 **Junior Editorial Officer**, International Journal of Artificial Intelligence in Education

COMMITTEE MEMBERSHIPS

2026 **Executive Committee Student Member**, Society for Learning Analytics Research (SoLAR)
2026 **PhD Admissions Committee**, Carnegie Mellon University
2025 **IJAIED Track Chair**, International Conference on Artificial Intelligence in Education (AIED)
2025 **Research Track Junior Advisory Board Member**, International Conference on Artificial Intelligence in Education (AIED)
2023-2026 **Graduate Student Assembly Representative**, Carnegie Mellon University
2023-2025 **NSF REU Summer Intern Admissions Committee**, Carnegie Mellon University

POLICY & ADVISORY CONTRIBUTIONS

2026 **OECD Digital Education Outlook 2026**, Pardos, Z. & Borchers, C., Chapter 11: *AI in Institutional Workflows: Learning from Higher Education to Unlock New Affordances for Education Systems and Institutions*, in *OECD Digital Education Outlook 2026: Exploring Effective Uses of Generative AI in Education* (OECD Publishing).
Institute of Education Sciences (IES), Co-author, Response to RFI on Education Research
2025 **Priorities** (Lead: J. Roschelle; with S. Ritter, A. Closser, C. Borchers). Submitted recommendations on leveraging large-scale research infrastructure through SEERNet.

CONFERENCE VOLUNTEERING

2023-2024 **EDM**, Student Volunteer
2023-2024 **ISLS**, Student Volunteer